GeoGebra-Based Mathematics Teaching Resources

Class 6 Mathematics Term – I

Chapter 2 – Introduction to Algebra

Teacher Documentation and Video Guide

Purpose: Support teachers in teaching Class 6 algebra concepts using GeoGebra-based visual demonstrations from the mathematics textbook.

Objectives of the Documentation

The objectives of this documentation are:

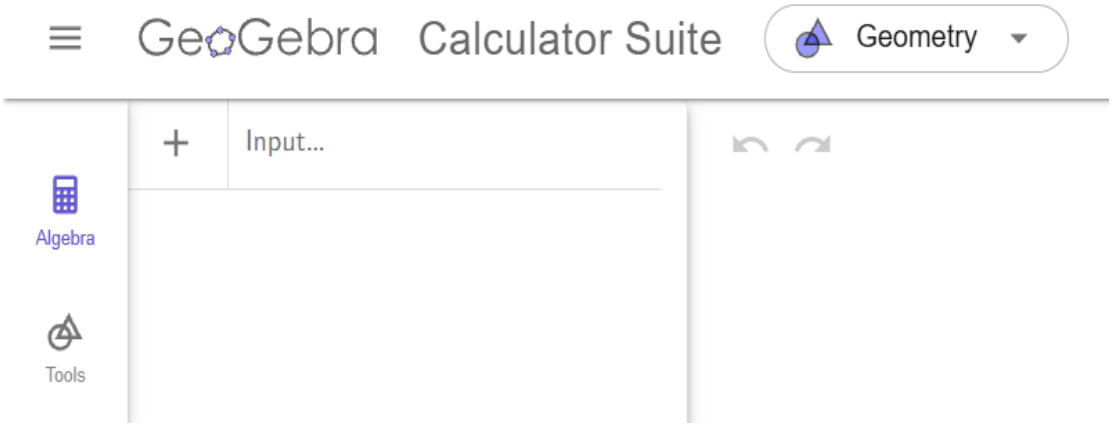
(i) To organise and document the GeoGebra instructional videos developed from Class 6 mathematics textbook problems in Chapter 2, linking each video to its relevant chapter, topic, and mathematical concept.

(ii) To guide teachers in using GeoGebra demonstrations effectively for concept explanation, classroom teaching, and lesson preparation.

GeoGebra Tools Used in the Activities

The GeoGebra activities were developed using various tools available in the GeoGebra Geometry interface. These tools help create interactive mathematical activities and support the visual representation of mathematical concepts.

1. Algebra Input

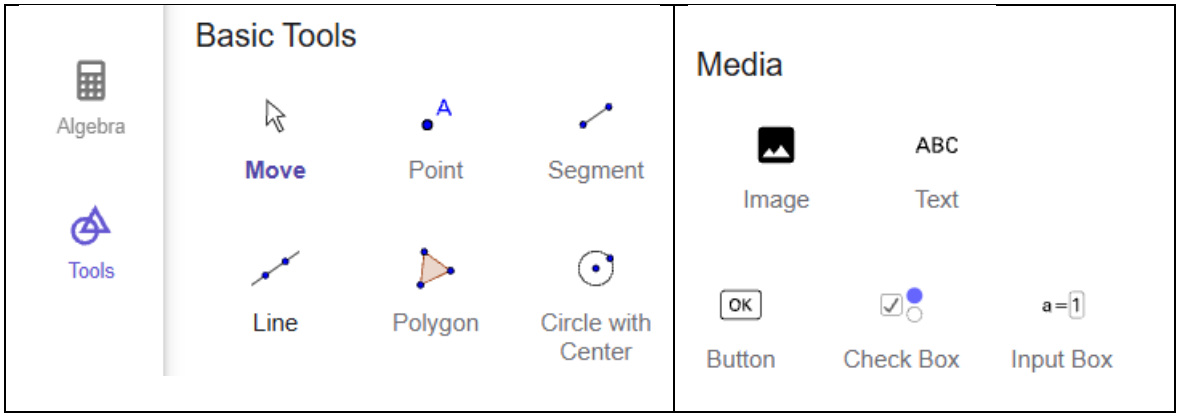


The Algebra Input field, labelled "Input…" in the GeoGebra Calculator Suite interface, is used to enter mathematical expressions, define variables, and execute commands. In activities, it defines numerical values, performs calculations, creates dynamic text, checks students' answers, and sets commands for interactive features such as the New Challenge button.

Procedure: Select the Algebra panel on the left side of the interface, click the "Input…" field, type the required expression or command, and press Enter to execute it.

2. Tools Options

- **Text Tool:** The Text tool lets you insert titles, mathematical questions, instructions, and explanations into the activity. It helps present the content clearly and organise the activity for learners.
- **Input Box Tool:** The Input Box tool allows students to enter their answers directly into the activity. It can be linked to a variable so that the entered value can be used for calculations and answer checking.



- **Button Tool:** The Button tool is used to perform specific actions when clicked. It can execute commands, reset values, generate new numbers, or provide a new set of questions for students to practise.
- **Image Tool:** The Image tool is used to insert pictures or visual elements into the activity. Images can make the activity more engaging and provide visual feedback, such as displaying a congratulatory image when students complete the activity successfully.
- **Segment Tool:** The Segment Tool is used to draw a line segment between two points. It helps teachers demonstrate line segments, measure lengths, and construct geometric figures for mathematical activities.
- **Regular Polygon Tool:** The Regular Polygon Tool is used to construct regular polygons by selecting two points and specifying the number of vertices. It helps teachers demonstrate the properties and shapes of regular polygons, such as equilateral triangles and squares.

Question 1 (Page 38 in the textbook)

Find the unknown.

(i) $37 + 43 = 43 + \square$ (ii) $(22 + 10) + 15 = \square + (10 + 15)$

(iii) If $7 \times 46 = 322$ then $46 \times 7 = \square$

Activity 1: Math Detective – Find the Unknown

Step 1: Create the activity title

Open GeoGebra Calculator Suite and select the geometry interface. From the Tools options, select the Text tool and create a new activity titled "MATH DETECTIVE: Find the Unknown!".

Step 2: Define the numbers used in the activity

Open the Input Bar in GeoGebra and enter the following values to define the variables used in the three questions.

$a = 37$; $b = 43$; $c = 22$; $d = 10$; $e = 15$; $f = 7$; $g = 46$

These variables represent the numbers used in the questions and let you update the values when a new challenge is generated.

Step 3: Calculate the product

Enter the following command in the Input Bar:

p = f * g

This command calculates the product of f and g.

Step 4: Create variables for student answers

Enter the following commands to create variables that will store the answers entered by students:

ans1 = 0; ans2 = 0; ans3 = 0

These variables are initially set to zero and are linked to the input boxes provided for the three questions.

Step 5: Create the questions using dynamic text

Use the Text command in the Input Bar to display the questions. Enter the following commands:

Question 1: Text("1. "+a+" + "+b+" = "+b+" +")

Question 2 — 1st part: Text("2. ("+c+" + "+d+") + "+e+" =")

Question 2 — 2nd part: Text(" + ("+d+" + "+e+")")

Question 3: Text("3. If "+f+" × "+g+" = "+p+", then "+g+" × "+f+" =")

“These commands display the questions using the defined variables. The second question is divided into two text objects so that an input box can be placed between them”.

Step 6: Insert input boxes

Select the Input Box tool from the GeoGebra toolbar and create three input boxes.

Link the input boxes to the following variables:

- First input box: ans1
- Second input box: ans2
- Third input box: ans3

These boxes allow students to enter their answers for the three questions.

Step 7: Create the answer-checking conditions

Enter the following commands in the Input Bar to check whether the answers entered by students are correct.

correct1 = ans1 == a

correct2 = ans2 == c

correct3 = ans3 == p

Each command compares the student's answer with the corresponding correct value. The result becomes true when the answer matches the expected value and false otherwise.

Step 8: Add automatic feedback messages

Enter the following commands to provide feedback based on the answers.

feedback1 = If(correct1,"Correct", "Try again") %For Question 1

feedback2 = If(correct2,"Correct", "Try again") %For Question 2

feedback3 = If(correct3,"Correct", "Try again") %For Question 3

The activity displays a positive message when the answer is correct and displays “Try again” when the answer is incorrect.

Step 9: Add explanations of the mathematical concepts

Use the Text tool to insert the following concept explanations into the activity.

For Question 1: Concept: Commutative property of addition.

For Question 2: Concept: Associative property of addition.
For Question 3: Concept: Commutative property of multiplication.
These explanations help teachers connect the textbook questions to the mathematical properties they demonstrate.

Step 10: Create the "New Challenge" button

Insert a button and name it New Challenge. Add the following commands to its scripting area:
SetValue(a, RandomBetween(10,50))
SetValue(b, RandomBetween(10,50))
SetValue(c, RandomBetween(10,50))
SetValue(d, RandomBetween(5,20))
SetValue(e, RandomBetween(5,20))
SetValue(f, RandomBetween(2,10))
SetValue(g, RandomBetween(10,50))
SetValue(ans1, 0)
SetValue(ans2, 0)
SetValue(ans3, 0)
The RandomBetween command generates new numbers within the specified ranges. The last three commands reset the answer variables to zero. When the teacher or student clicks the button, a new set of questions can be attempted.

Step 11: Set the final success condition

Enter the following command:
allCorrect = correct1 && correct2 && correct3
This command combines the three answer-checking conditions. The variable allCorrect becomes true only when all three answers are correct.

Step 12: Insert the celebration image

Insert the "Well Done" image into the GeoGebra activity.

1. Click the image option in the Tools menu.
2. Browse and select your own well-done image.
3. Open Advanced settings.
4. Locate Condition to Show Object under the visibility settings.
5. Enter the following condition: allCorrect in visibility settings.
6. For the ideal position, set (3,-7), (9,-7) and (3,-3)

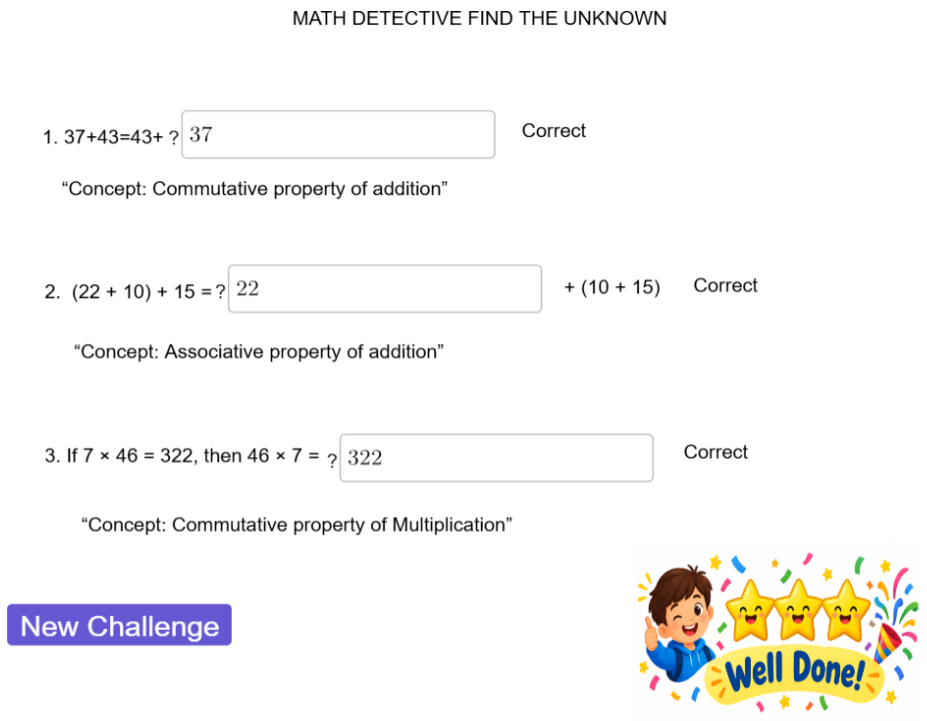
The image remains hidden while the questions are being solved and appears automatically when all three answers are correct.

Step 13: Test the completed activity

Run the activity and verify the following:

1. All three questions are displayed correctly.
2. Students can enter answers using the input boxes.
3. Incorrect answers display the "Try again." message.
4. Correct answers display the appropriate feedback message.
5. The New Challenge button generates new numbers and resets the answer variables.
6. The "Well Done" image appears after all three answers are correct.

Final Output of Activity 1



Question 2 (Page 42 in the textbook)

Complete the following pattern.

$$9 - 1 =$$
$$98 - 21 =$$
$$987 - 321 =$$
$$9876 - 4321 =$$
$$98765 - 54321 =$$

What comes next?

Activity 2: Math Detective – Discover the Number Pattern

Step 1: Create the activity title

Open GeoGebra Calculator Suite and select the geometry interface. From the Tools options, select the Text tool and create the activity title: MATH DETECTIVE: Discover the number pattern

Step 2: Create the number pattern questions

Select the Algebra panel, click the Input… field, and enter the following commands one by one to display the five subtraction questions.

Question 1: Text("1. 9-1=")
Question 2: Text("2. 98-21=")
Question 3: Text("3. 987-321=")
Question 4: Text("4. 9876-4321=")

Question 5: Text("5. 98765-54321=")
These commands display the subtraction questions in the activity for students to calculate and observe the number pattern.

Step 3: Create variables for student answers

Enter the following commands in the Input field:
ans1=0; ans2=0; ans3=0; ans4=0; ans5=0
These variables store the answers entered by students for Questions 1–5. Each variable is initially set to zero.

Step 4: Create the answer-checking conditions

Enter the following commands to check the answers for the five questions:
c1=ans1==8
c2=ans2==77
c3=ans3==666
c4=ans4==5555
c5=ans5==44444
Each condition compares the student's answer with the expected answer. The condition becomes true when the entered answer matches the expected value.

Step 5: Add feedback messages for the first five questions

Enter the following commands to display feedback based on the answer-checking conditions.
Question 1: f1=Text(If(c1,"Correct!","Try Again!"))
Question 2: f2=Text(If(c2,"Correct!","Try Again!"))
Question 3: f3=Text(If(c3,"Correct!","Try Again!"))
Question 4: f4=Text(If(c4,"Correct!","Try Again!"))
Question 5: f5=Text(If(c5,"Correct!","Try Again!"))
These commands display “Correct!” when the answer is correct and “Try Again!” when the answer is incorrect.

Step 6: Display the sixth question

Enter the following command in the Input field: Text("6. What Comes Next?")
This command displays the sixth question, which encourages students to observe the results of the previous subtraction questions and identify the next number pattern.

Step 7: Create the answer variable for Question 6

Enter the following command:
ans6=0
This variable stores the answer entered by the student for Question 6.

Step 8: Create the answer-checking condition for Question 6

Enter the following command:
c6=ans6==987654-654321
This command checks whether the answer entered by the student matches the result of the subtraction expression 987654-654321.

Step 9: Add feedback for Question 6

Enter the following command:
f6=Text(If(c6, "Correct! Excellent detective work", "Study the answers carefully: Look at the repeated digits in the answer!"))

This command displays “Correct! Excellent detective work” when the answer is correct. Otherwise, it displays a hint encouraging students to study the repeated digits in the answers.

Step 10: Display the answer prompt for Question 6

Enter the following command:

```
Text("Enter your answer for Q6:")
```

This command displays a prompt asking students to enter their answer for Question 6.

Step 11: Create the additional answer variable

Enter the following command:

```
ans7=0
```

This variable stores an additional answer for the activity.

Step 12: Create the seventh answer-checking condition

Enter the following command:

```
c7=ans7==333333
```

This command checks whether the value stored in ans7 is equal to 333333.

Step 13: Add feedback for the seventh answer

Enter the following command:

```
f7=Text(If(c7,"Fantastic!","Try Again"))
```

This command displays “Fantastic!” when the answer is correct and “Try Again” when the answer is incorrect.

Step 14: Create the overall completion condition

Enter the following command:

```
allCorrect=c1 && c2 && c3 && c4 && c5 && c6 && c7
```

This command combines all seven answer-checking conditions. The variable allCorrect becomes true only when all seven conditions are true.

Step 15: Insert and link the input boxes

Select the Input Box tool from the Tools options and create input boxes for the student answers. Link the boxes to the corresponding answer variables: ans1, ans2, ans3, ans4, ans5, ans6, and ans7.

Arrange the input boxes alongside their respective questions so students can enter their answers easily.

Step 16: Insert the “Well Done” image

Select the Image tool from the GeoGebra Tools options and insert the “Well Done” image into the activity. Position the image appropriately so that it is visible when students complete all the questions correctly.

Step 17: Set the visibility condition for the image

1. Select the inserted “Well Done” image.
2. Open the Settings panel and locate the Advanced options.
3. Find Condition to Show Object under the visibility settings.
4. Enter the following condition: allCorrect

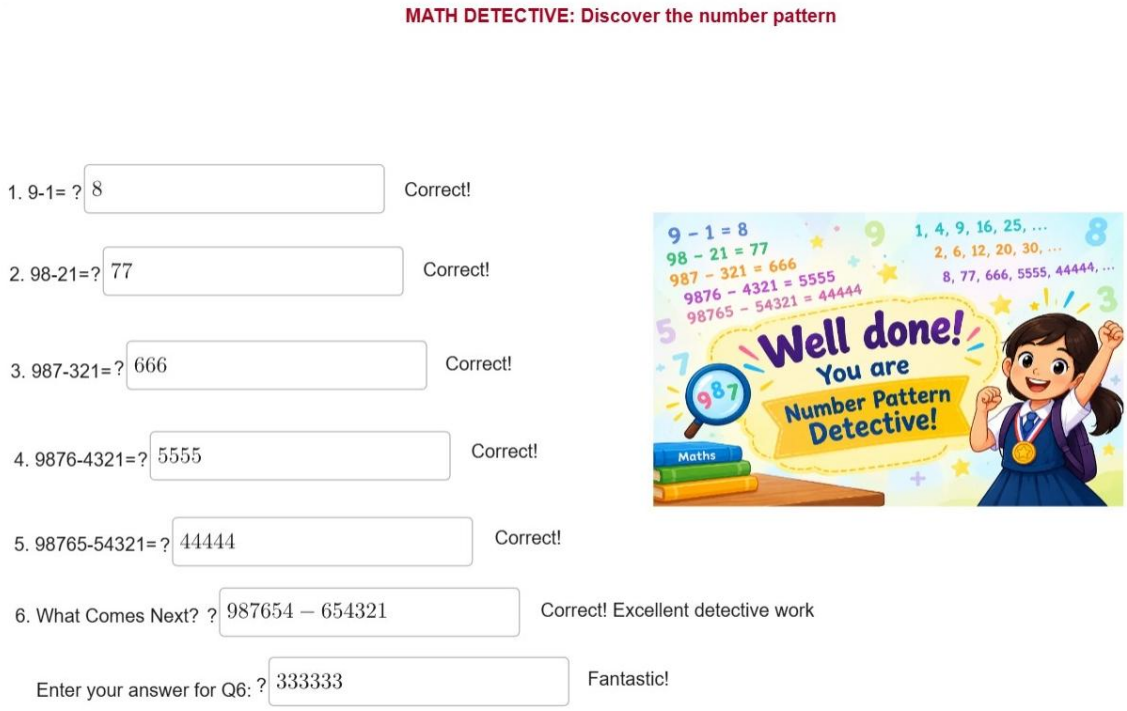
The image will remain hidden while students are answering the questions and will appear when all seven answer-checking conditions are true.

Step 18: Test the completed activity

Run the activity and verify that:

1. All questions and input boxes are displayed correctly.
2. The feedback messages appear according to the answers entered.
3. The "Well Done" image remains hidden until all seven answers are correct.
4. The "Well Done" image appears automatically when all seven answer-checking conditions are true.

Final Output of Activity 2



Question 3 (Page 42 in the textbook)

A piece of wire is '12s' cm long. What will be the length of the side, if it is formed as

(i) an equilateral triangle

(ii) a square

Activity 3: Finding the Side Length of an Equilateral Triangle and a Square

Step 1: Display the question

Open GeoGebra Calculator Suite and select the geometry interface. Use the Text tool to display the following question without changing its content.

A piece of wire is 12 cm long. What will be the length of the side if it is formed as

(i) An equilateral triangle

(ii) (ii) A square

Step 2: Create the geometric shapes

Select the Segment Tool from the Tools options to create line segments. Use the Regular Polygon Tool to construct an equilateral triangle and a square, representing the two shapes formed using the wire.

Step 3: Define the wire length

In the Algebra panel, click the Input… field and enter:

L = 12

This command defines the variable L as the wire length used in the GeoGebra activity.

Step 4: Calculate the side length of the equilateral triangle

Enter the following command in the Input field:

e = ((L)/(3))

This command calculates the side length of the equilateral triangle by dividing the wire length by 3, since an equilateral triangle has three equal sides.

Step 5: Calculate the side length of the square

Enter the following command:

s = ((L)/(4))

This command calculates the side length of the square by dividing the wire length by 4, since a square has four equal sides.

Step 6: Display the wire length

Enter the following command:

Text("Wire length = "+L+" cm")

This displays the current wire length in the activity.

Step 7: Display the triangle side length

Enter the following command:

Text("Triangle Side= "+e+" cm")

This displays the calculated side length of the equilateral triangle.

Step 8: Display the square side length

Enter the following command:

Text("Square Side= "+s+" cm")

This displays the calculated side length of the square.

Step 9: Create a button to change the wire length

Select the Button Tool from the Tools options and create a button to generate a different wire length. Set the button's script to:

SetValue(L, RandomBetween(4,30))

When the button is clicked, a new value for L is generated between 4 and 30.

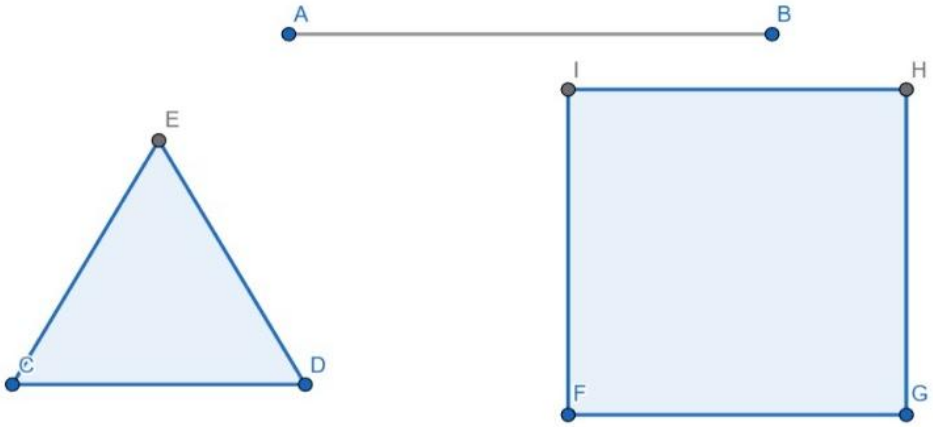
Step 10: Test the completed activity

Run the activity and verify that:

1. The question and both geometric shapes are displayed.
2. The wire length is shown correctly.
3. The side lengths of the equilateral triangle and square are calculated and displayed.
4. Clicking the button generates a different wire length and updates the corresponding calculations.

Final Output of Activity 3

A piece of wire is 12 cm long. What will be the length of the side, if it is formed as
(i) An equilateral triangle
(ii) A square



Wire length = 12 cm

Triangle Side= 4 cm

Square Side= 3 cm

Different Length of the string

Question 4 (Page 42 in the textbook)

Identify the value of the shapes and figures in the table given below and verify their addition horizontally and vertically.

				= 30
				= 36
				=
				=
= 32	=	=	=	=

Activity 4: Shape Detective – Find Each Shape's Value

Step 1: Create the activity title

Open GeoGebra Calculator Suite and select the geometry interface. From the Tools options, select the Text tool and create the following title and subtitle:
SHAPE DETECTIVE and Find each shape's value

Step 2: Display the shape-value questions and hints

Use the Text tool to display the following questions and their corresponding hints, as given in the activity code.
Text("1. Value of Star") Text("Hint: 32/4")
Text("2. Value of Smiley") Text("Hint: (30 – 16)/2")
Text("3. Value of Triangle") Text("Hint: (36 – 16)/2")
These questions encourage students to use the hints to determine the shapes' values.

Step 3: Create variables for the answers

In the Algebra panel, click the Input… field and enter the following commands:
ans1 = 0; ans2 = 0; ans3 = 0
These variables store the answers entered by students for the three shape-value questions.

Step 4: Create the answer-checking conditions

Enter the following commands to check the answers:
c1 = ans1 == 8
c2 = ans2 == 7
c3 = ans3 == 10
Each condition compares the student's answer with the expected value for each shape.

Step 5: Add feedback messages

Use the Text tool and the If command to display feedback for each answer.
Text(If(c1, "Correct!", "Try again"))
Text(If(c2, "Correct!", "Try again"))
Text(If(c3, "Correct!", "Try again"))
The activity displays “Correct!” when the answer is correct and “Try again” otherwise.

Step 6: Display the missing-total questions

Use the Text tool to display the heading and these questions:
Text("Find the missing totals")
Text("4. Row 3 Total = ")
Text("5. Row 4 Total =")
Text("6. Column 2 Total =")
Text("7. Column 3 Total =")
Text("8. Column 4 Total =")
These questions require students to calculate the specified row and column totals from the shape-value puzzle.

Step 7: Create variables for the missing totals

Enter the following commands in the Algebra Input field:
ansr3 = 0; ansr4 = 0; ansCol2 = 0; ansCol3 = 0; ansCol4 = 0
These variables store the answers entered for the row and column total questions.

Step 8: Create the answer-checking conditions for the totals

Enter the following commands:
cR3 = ansr3 == 32
cR4 = ansr4 == 32
cl2 = ansCol2 == 32
cl3 = ansCol3 == 34
cl4 = ansCol4 == 32
These conditions check whether the entered row and column totals match the expected values defined in the activity code.

Step 9: Add feedback messages for the totals

Enter the following commands to provide feedback for each question:
Text(If(cR3, "Correct!", "Try again"))
Text(If(cR4, "Correct!", "Try again"))
Text(If(cl2, "Correct!", "Try again"))
Text(If(cl3, "Correct!", "Try again"))
Text(If(cl4, "Correct!", "Try again"))
Each message provides immediate feedback based on the corresponding answer-checking condition.

Step 10: Display the final-answer section

Use the Text tool to add the heading and display the following questions:
Text("Find the final answer")
Text("Sum of Row Totals =")
Text("Sum of Column Totals =")These questions encourage students to calculate the sum of the row totals and the sum of the column totals.

Step 11: Create variables for the final answers

Enter the following commands in the Algebra Input field: ansrow = 0; anscol = 0
These variables store the answers entered by students for the sum of the row totals and the sum of the column totals.

Step 12: Create the final answer-checking conditions

Enter the following commands:
crow = ansrow == 130; ccol = anscol == 130
These conditions check whether both final answers are equal to 130.

Step 13: Add feedback for the final answers

Enter the following commands:
Text(If(crow, "Correct!", "Try again"))
Text(If(ccol, "Correct!", "Try again"))
The activity displays the appropriate feedback depending on whether each final answer is correct.

Step 14: Create the overall completion condition

Enter the following command:
allCorrect = c1 && c2 && c3 && cR3 && cR4 && cl2 && cl3 && cl4 && crow && ccol
This command combines all ten answer-checking conditions. The variable allCorrect becomes true only when all three shape values, all five row and column totals, and both final answers are correct.

Step 15: Insert and link the input boxes

Select the Input Box tool and create input boxes for the answers. Link each box to its corresponding variable:

- Shape values: ans1, ans2, ans3
- Row and column totals: ansr3, ansr4, ansCol2, ansCol3, ansCol4
- Final answers: ansrow, anscol

Arrange the boxes beside their respective questions.

Step 16: Insert the Completion Image

Insert the "Well Done" image using the Image tool. Under **Advanced Settings → Condition to Show Object**, enter the appropriate completion condition for the activity. Adjust the image position as required. The image appears automatically when all the answers are correct.

Step 17: Test the completed activity

Run the activity and check that:

1. The shape-value questions and hints are displayed correctly.
2. Students can enter their answers in the input boxes.
3. The row and column total questions are displayed correctly.
4. The correct feedback messages appear for each answer.
5. Both final-answer questions accept the expected value of 130.
6. The allCorrect condition becomes true when all ten answers are correct.
7. The "Well Done" image remains hidden until all the answers are correct.
8. The "Well Done" image appears automatically when all the answer-checking conditions are true.

Final Output of Activity 4

SHAPE DETECTIVE

Find Each Shape Value

1. Value of Star= ? 8 Correct! Hint : 32/4
2. Value of Smiley= ? 7 Correct! Hint : ((30 – 16)/(2))
3. Value of Triangle= ? 10 Correct! Hint: ((36–16)/(2))

Find the Missing Totals

4. Row 3 Total = ? 32 Correct!
5. Row 4 Total = ? 32 Correct!
6. Column 2 Total = ? 32 Correct!
7. Column 3 Total = ? 34 Correct!
8. Column 4 Total = ? 32 Correct!



Find the Final Answer

Sum of Row Totals =? 130 Correct!

Sum of Column Totals = ? 130 Correct!